

Exercise 11

```
main.c
1 #include <stdio.h>
2 int e[10][10], visited[10], n;
3 void eclosure(int state)
4 {int i;
5     visited[state] = 1;
6     printf("q%d ", state);
7     for(i = 0; i < n; i++)
8     {if(e[state][i] == 1 && !visited[i])
9         eclosure(i);
10    }
11 }
12 int main(){int i, j;
13     printf("Enter number of states: ");
14     scanf("%d", &n);
15     printf("Enter epsilon transition matrix:\n");
16     for(i = 0; i < n; i++)
17     {for(j = 0; j < n; j++)
18         scanf("%d", &e[i][j]);
19     }
20     printf("\nEpsilon Closures:\n");
21     for(i = 0; i < n; i++)
22     {for(j = 0; j < n; j++)
23         visited[j] = 0;
24         printf("E-Closure(q%d): ", i);
25         eclosure(i);
26         printf("\n");
27     }
28     return 0;
29 }
```

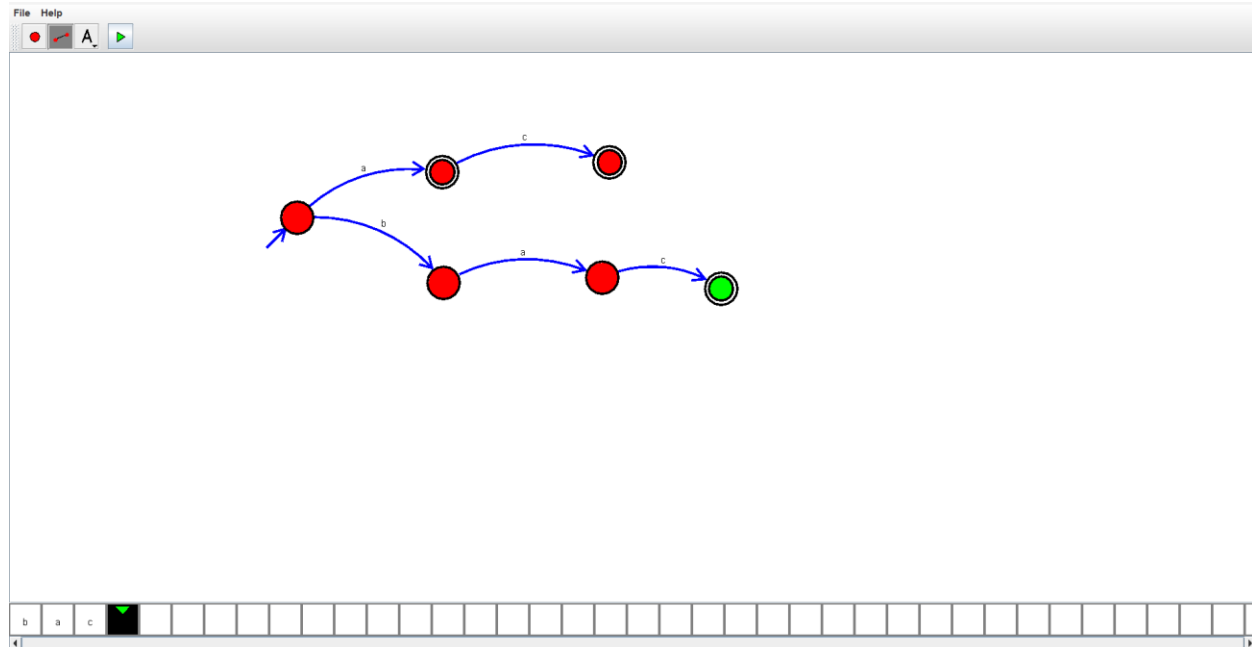
Output

Enter number of states: 4
Enter epsilon transition matrix:
0 1 0 0
0 0 1 0
0 0 0 1
0 0 0 0

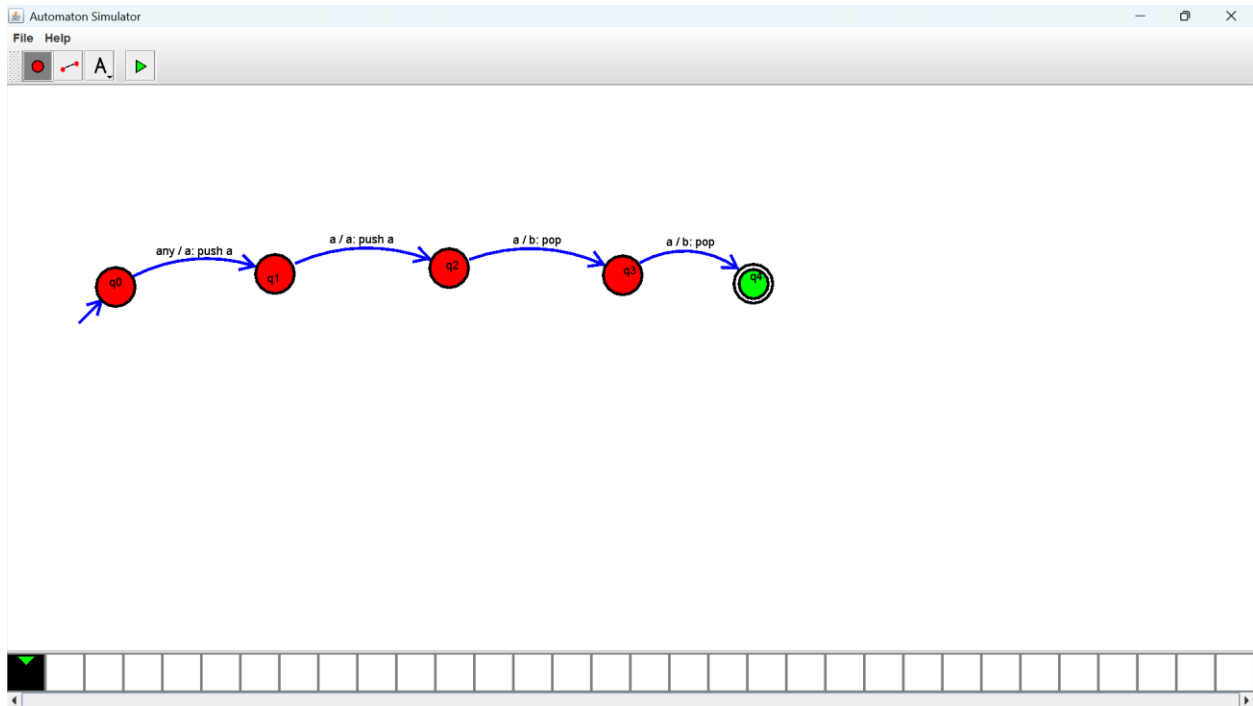
Epsilon Closures:
E-Closure(q0): q0 q1 q2 q3
E-Closure(q1): q1 q2 q3
E-Closure(q2): q2 q3
E-Closure(q3): q3

=== Code Execution Successful ===

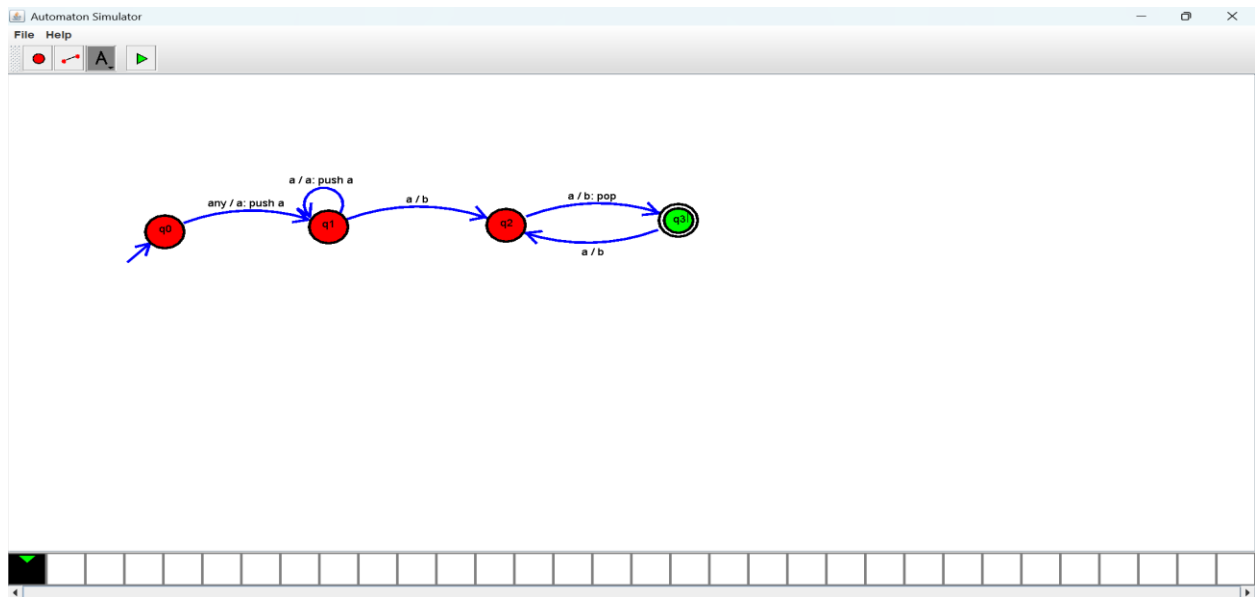
Exercise 12



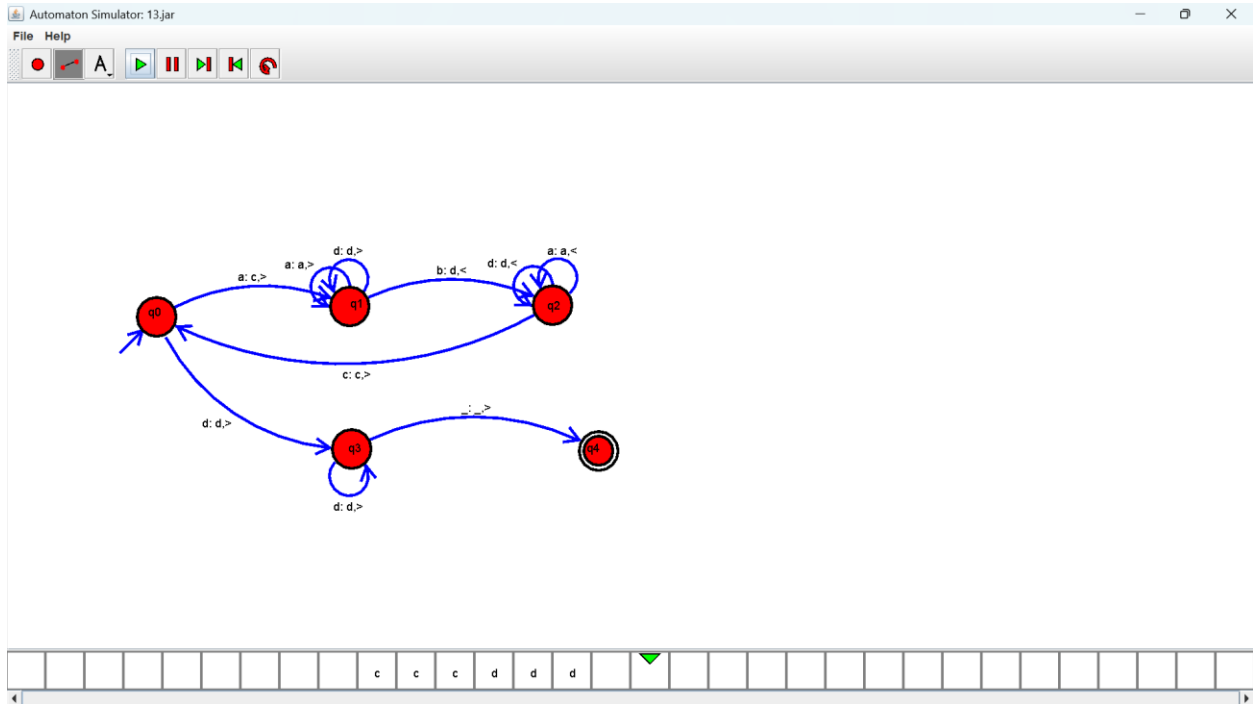
Exercise 13



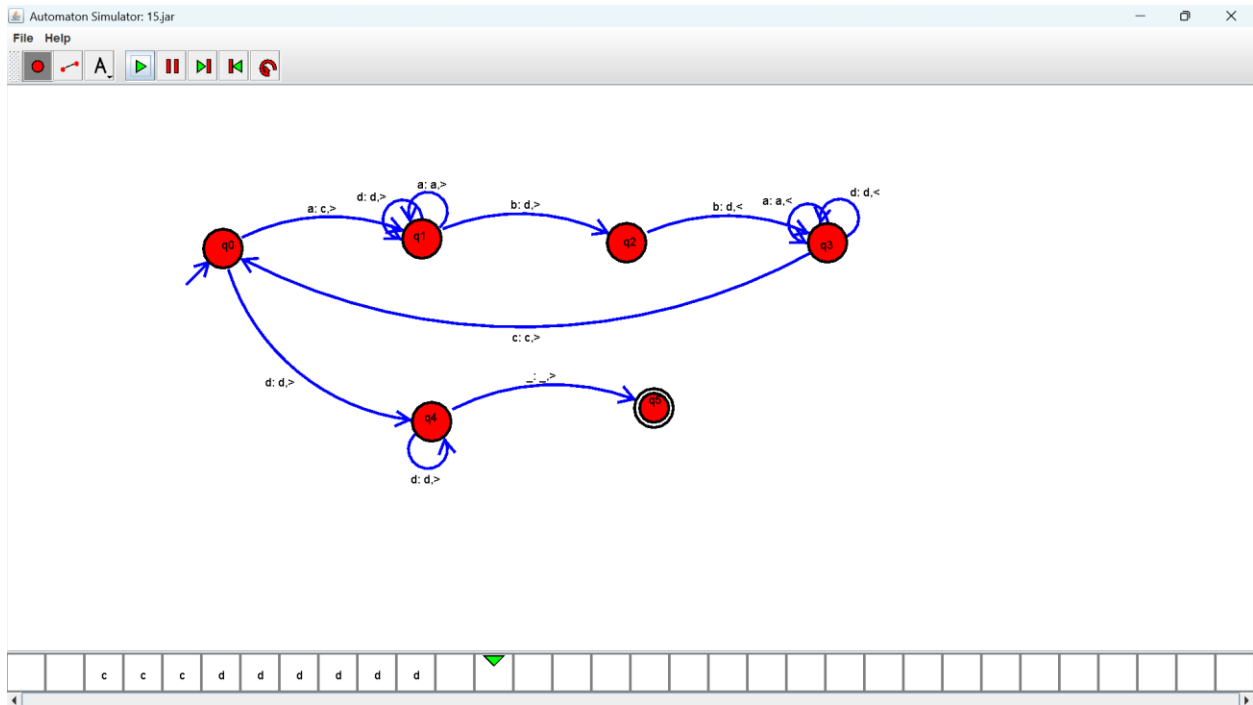
Exercise 14



Exercise 15

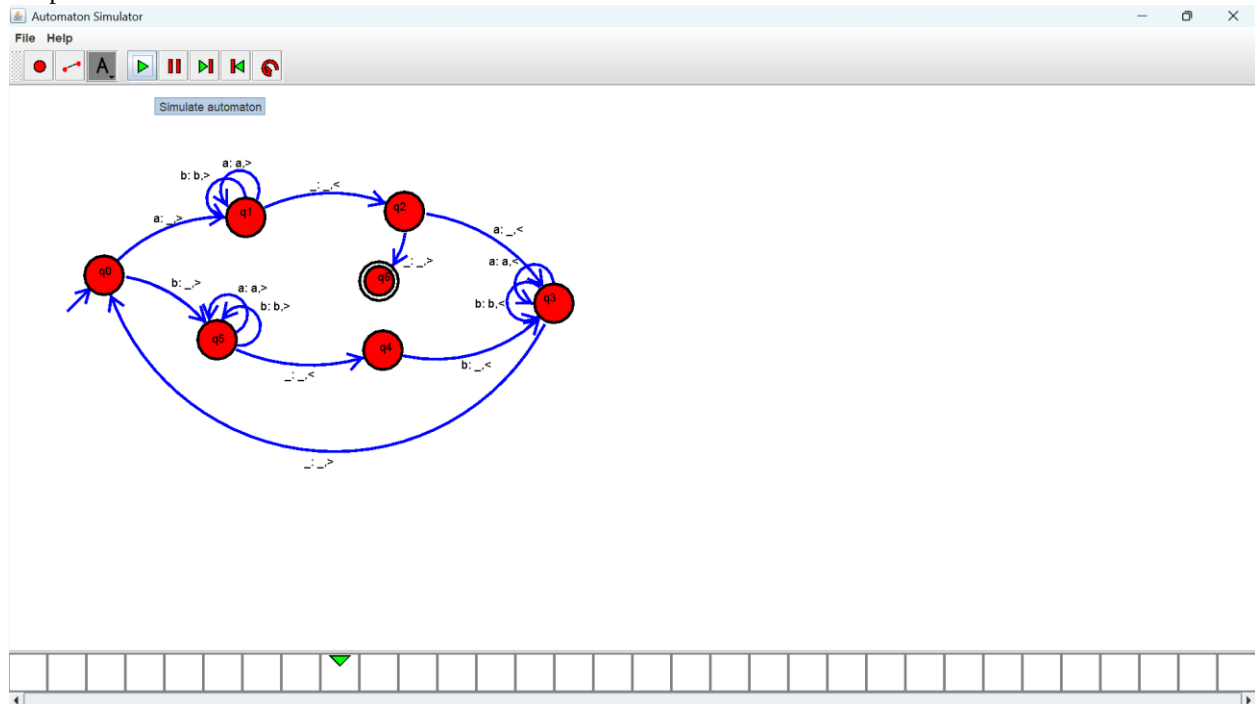


Exercise 16

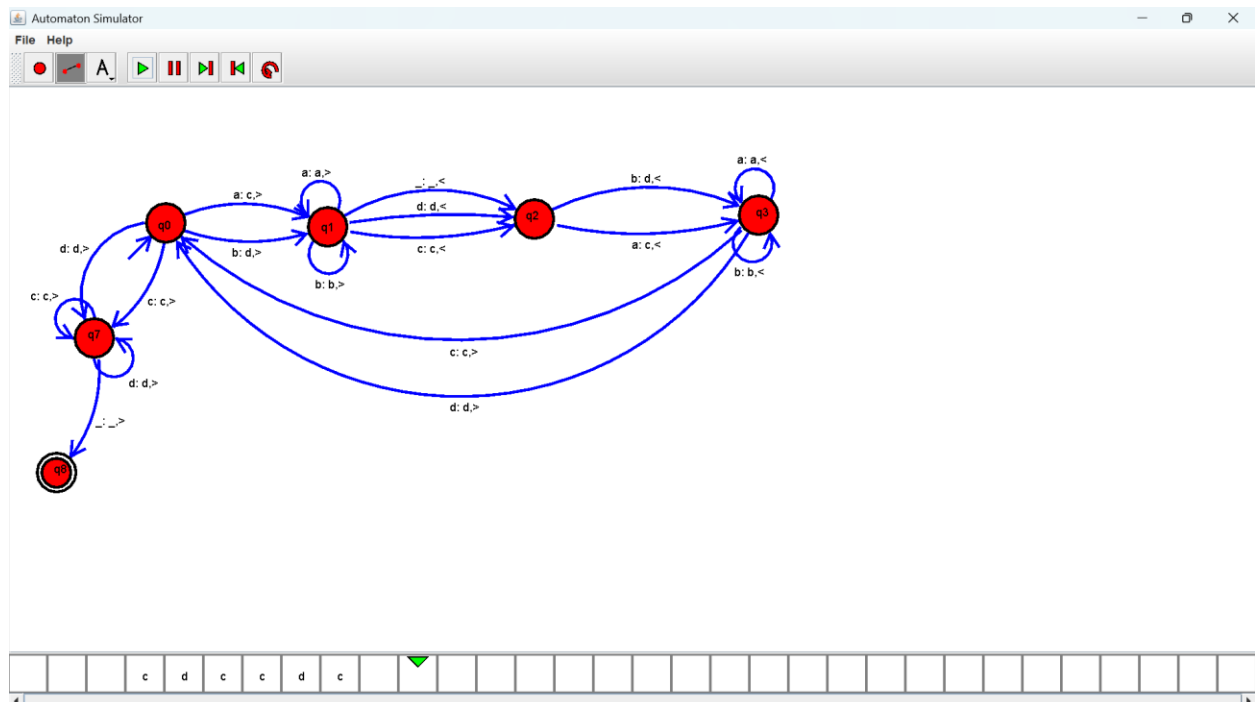


Exercise 17

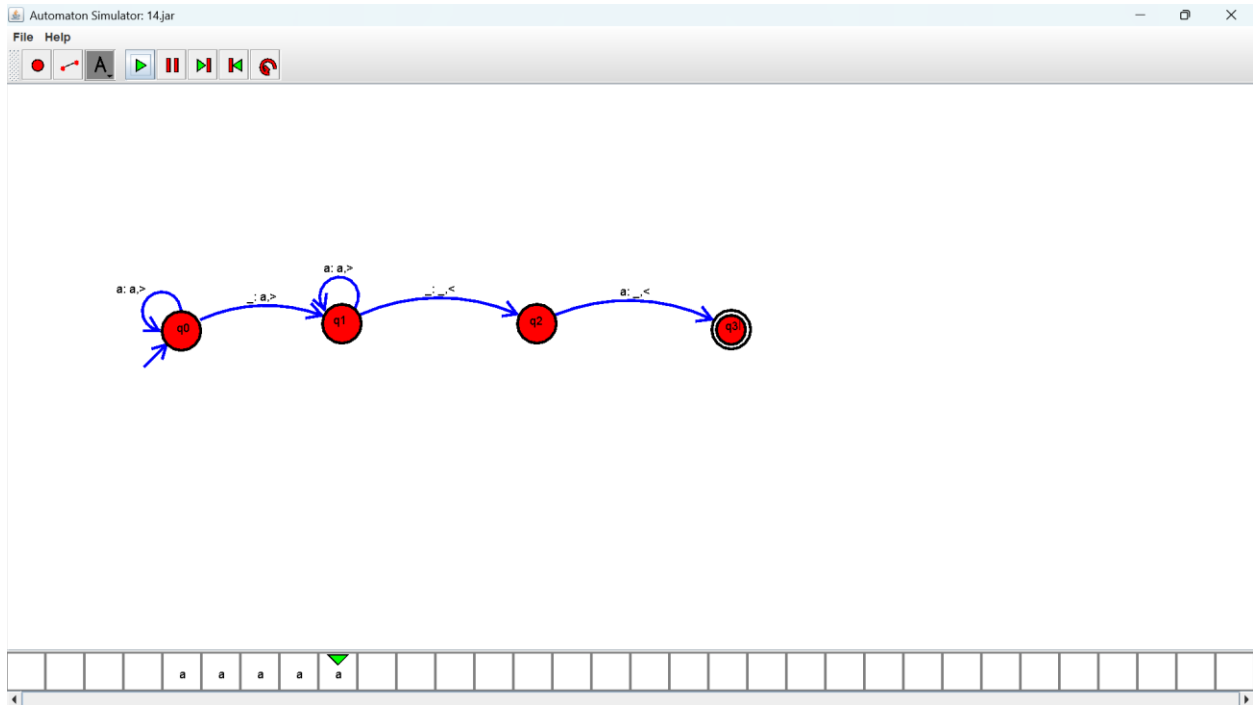
Output :



Exercise 18



experimnet 19



Exercise 20

